

Quiz 1: KKT Regimes in a Quantity-Quality Household

ECON 803 - Microeconomic Theory I - Fall 2026

October 2, 2026 - 25 minutes - 20 points

Name: Student number:

Show the KKT conditions and enough algebra to identify every active set. Children are divisible in this classroom model.

From Galor's mechanism to three static snapshots

In Galor's Unified Growth Theory, the time cost of n children with education e is proportional to $n(\tau + e)$. A higher return to education moves resources from child quantity toward child quality. Galor's model is dynamic and maintains positive fertility. To study three KKT active sets in one class, we use the following static teaching extension.

Let $H \geq 0$ be total purchased human capital for all children. If $n > 0$, quality per child is $q = H/n$. Each regime also specifies a small fertility floor $\underline{n}_r \geq 0$. The household chooses $c, H \geq 0$ and $n \geq \underline{n}_r$ to maximize

$$U(c, n, H) = c + 6n - \frac{1}{2}n^2 + 4H - \frac{1}{2}H^2$$

subject to the regime-specific budget and quality cone

$$c + pn + kH + \frac{1}{2}H^2 \leq 20, \quad sn \leq H \leq 4n, \quad n \geq \underline{n}_r.$$

The term $H^2/2$ makes total education expenditure convex. The cone makes $q \in [s, 4]$ when children are present and forces $H = 0$ when a regime permits $n = 0$.

1. General KKT system [8 points]

- (a) Use multipliers $\lambda, \mu_c, \mu_n, \alpha, \beta \geq 0$ for, respectively, the budget slack, $c \geq 0$, $n - \underline{n}_r \geq 0$, $H - sn \geq 0$, and $4n - H \geq 0$. Write the Lagrangian and the complete KKT system. [6]
- (b) Explain why these conditions are sufficient. Show that whenever $c^* > 0$, the budget binds and $\lambda^* = 1$. [2]

2. Solve three regimes [9 points]

For each regime, find (c^*, n^*, H^*) , report $q^* = H^*/n^*$ when it is defined, identify the active inequalities, and give one valid multiplier vector. You may first use the binding budget to eliminate consumption. [3 points each]

Malthusian: $(p, k, s, \underline{n}_r) = (2, 5, 0, 0)$.

Modern quantity-quality balance: $(p, k, s, \underline{n}_r) = (4, 2, 0, 0)$.

High-quality-cost, low-fertility (post-modern) extension: $(p, k, s, \underline{n}_r) = (4, 6, 2, 1/4)$.

3. Interpretation [3 points]

- (a) Describe the movement from quantity to quality across your first two solutions. [1]
- (b) Explain why a high quality price k alone need not eliminate fertility, but a high k together with the minimum-quality requirement $H \geq 2n$ can push fertility down to its small lower bound. [1]
- (c) State one dynamic feedback in Unified Growth Theory that this quiz holds fixed. Also state why the binding low-fertility floor should be called our extension rather than Galor's prediction. [1]

Source note: inspired by Oded Galor, "From Stagnation to Growth: Unified Growth Theory," Handbook of Economic Growth 1A (2005), 171–293, especially equations (4)–(10). DOI: 10.1016/S1574-0684(05)01004-X. The quadratic objective, convex monetary education cost, quality cone, and binding low-fertility floor are teaching extensions.